

Julian Stoller

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EDUCATION

Cornell University, College of Engineering

Bachelor of Science in Mechanical Engineering

Ithaca, NY

May 2026

Relevant Courses: Spaceflight Mechanics, Intermediate Fluid Dynamics, Additive Manufacturing, Product Design

EXPERIENCE

Mechanical Engineering Intern

LLUME | Ithaca, NY

June 2026 – July 2026

- Created and analyzed dynamic, multi-body, impact simulations to identify structural weaknesses in SLA printed parts and the overall assembly using Fusion 360's simulation workspace
- Designed and prototyped consumer-oriented mechanisms for a wearable biomedical device
- Sourced custom hardware from machine shops; worked with suppliers to make our parts easier to fabricate
- Attending daily sync meetings and presented work in group design reviews

Mechanical Engineering Intern

InocuJect

April 2023 – August 2025

- Prototyped electromechanical systems intended for use in the medical device industry, creating 3D printed prototypes using CAD and engineering analysis principles to meet customer and product specifications
- Designed quality control and inspection workflows for small-batch production using Six Sigma principles
- Developed sub-millimeter-scale manufacturing processes using CNC milling, custom tooling, and fixturing; authored work instructions and process documentation for new and ongoing fabrication methods
- Performed structural and lifecycle analysis on load bearing components, resulting in alternative design choices
- Troubleshoot circuits and control software to enable faster sensor communication, reducing latency up to 60% and improving production test throughput

Project Team Manager

Cornell Autonomous Bicycle | Ithaca, NY

February 2021 – April 2024

- Directed a cross-functional team of 30+ mechanical, electrical, and software engineers; ran design reviews and root-cause problem-solving sessions involving subject matter experts
- Designed and built several components used in an autonomous robot bicycle and a quadrupedal robot using ANSYS, Fusion360, and fabrication tools, assessed available manufacturing methods to drive production
- Reduced part cost by 98% by transitioning from vendor outsourcing to in-house manufacturing, applying DFM/DFA principles to improve producibility
- Lowered pre-production gearbox cost by 62% via design for sheet metal stamping, and to under 12% of original price while maintaining durability thresholds

Research Assistant

Architectural Robotics Lab | Ithaca, NY

June 2022 – April 2024

- Created specialized software to model the success of different materials used in custom components, resulting in optimization of material selection and component shape for parts in a custom robotic arm
 - Simulated robot locomotion dynamics in MATLAB and verified kinematic mechanisms using Fusion 360 assembly analysis to inform component design decisions and drive iterative design choices
 - Collected and analyzed data from human-robot interaction (HRI) studies and conducted participant interviews
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PROJECTS

Low-Cost Cycloidal Gear Box

Spring 2023

- Designed a cycloidal gearbox in SolidWorks to meet cost and performance specifications
- Collaborated with manufacturing services to lower tolerance and backlash using GD&T
- Prototyped and iteratively improved performance leading to a 23% increase in torque capacity

Rapidly Deployable Concentrated Solar Plant

Fall 2024

- Designed a concentrated solar energy plant which could be manufactured through AM methods
- Won 1st place among 81 entries in the Society of Manufacturing Engineers 2025 Digital Manufacturing Contest, winning \$1,000 and free access to the Rapid + TCT Additive Manufacturing Event.

- Authoring a provisional patent on the technology explored in the project's final report (ongoing)
- Series Elastic Actuator Gripper** Summer 2024
- Designed a set of variable gain rotary series elastic actuators to be used in a motorized gripper, allowing for safe human-compatible uses and industrial applications in the same package
 - Developed control software intended to dynamically change the operational regime in response to user input
 - Built and bench-tested a working prototype, demonstrating desired force control

HONORS & CERTIFICATIONS

- Six Sigma Green Belt Certification, Patent Holder (Pending Co-inventor Of Novel Fluid Delivery System), 2025 SME Digital Manufacturing Contest Winner, Big Red Hacks 2025 Finalist, Big Idea Competition Finalist

SKILLS

- Engineering Tools (AutoCAD, Fusion360, SolidWorks, ANSYS, PCB Design in KiCAD)
- Programming (MATLAB, Python, R, LabVIEW, Arduino, C++, Simulink)
- Manufacturing (CNC Milling, Machining, 3D Printing, Band Saw, SLA/FDM Printing, Rapid Prototyping, DFMA)
- Design & Analysis (Microsoft Office, CAD, FEA, GD&T, DFM, DFA, DFT, DFC, DFAM)